TIC 165227846

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

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BLS EPHEMERIS
  Period       2.096633 d  (ALIASED, see below)
    -> corrected period: 4.193266 d
  Epoch (t0)   3748.7797 BTJD
  Duration     0.0400 d
  Depth        9.897 %  (98,967 ppm)
  SNR          173.4

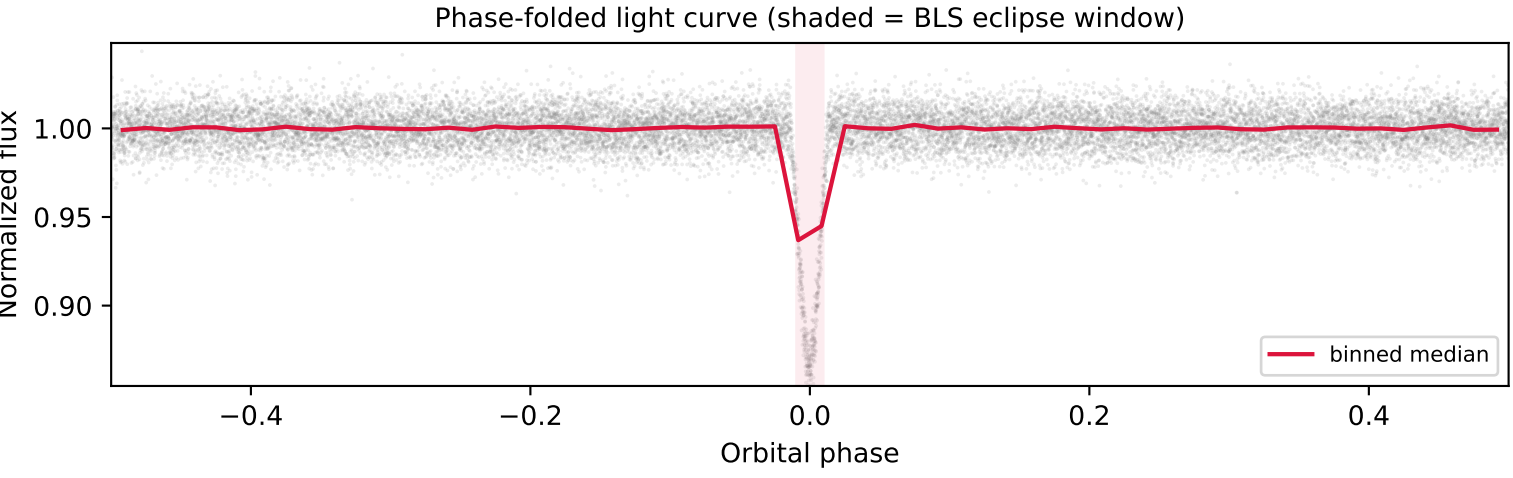
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     12  (gate >= 3)
  Robust depth ratio    1.02  (gate >= 0.50)
  Robust depth sigma    147.5  (gate >= 7.0)
  Odd/even z-score      6.79  (ALIAS FLAGGED)
  Secondary eclipse     not detected
```

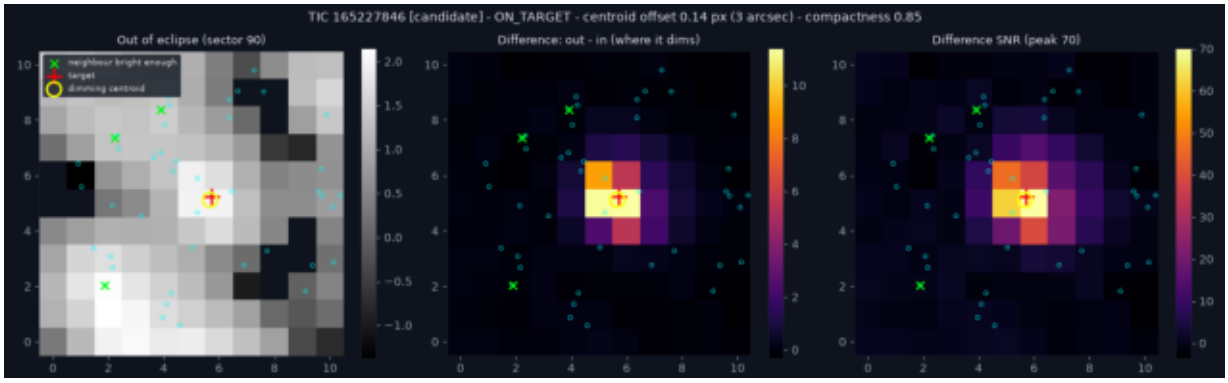
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 90    light-curve dip in this sector: 143.9 sigma    t0 refined by 0.0000 phase
  Difference-image peak SNR   70.15   (gate >= 5.0)      Compactness   0.85   (gate >= 0.50)
  Dimming centroid offset     0.14 px (3 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 165227834
    3.70 px away, dTmag +2.21 - farther from the dimming than the target
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Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 6.79 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 4.193266 d, in any write-up.